DAY 25 | PRE-READ

Small Models, Real Savings

A light read before Day 25. About 15 minutes, no code required.

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Why read this?

So far you have called one large model for everything. Today you learn that a small model is often enough, and how to work out what each AI feature really costs.

1. Not every job needs an expert NEW (v4)

A hospital receptionist can sort most visitors without a doctor. A **small language model (SLM)** is like that receptionist: it is small enough to run on a laptop or your own server, and it is very good at narrow jobs such as sorting tickets into categories.

2. Open-weight models NEW (v4)

Some companies publish the weights of their models so anyone can download and run them. Families you will hear about include Phi, Gemma, Llama, Qwen and Mistral's small models. Each has its own licence, so always check it before using one for a customer.

3. Packing lighter NEW (v4)

Quantisation stores each number in the model with fewer bits, like compressing a photo. The model takes less memory and runs faster, with a small loss in quality.

4. Paying by the word piece NEW (v4)

API models charge per token. What you send is **input**, what comes back is **output**, and output costs more. If you send the same long instructions every time, a **cache** can make the repeated part cheaper.

Everyday idea	AI term
Small, local helper	SLM
Compressing the model	Quantisation
Remembering the repeated part	Prompt caching
Ask the expert only when needed	Model routing
A spending limit	Budget or quota

Key idea: the cheapest token is the one you never send. Measure first, then cut.

5. Think about these before class

- Which SmartDesk jobs are simple enough for a small model?

- If 50,000 tickets arrive a day, why does saving a fraction of a cent per ticket matter?
- What customer data should never leave the company network?

Kenyaku (倹約): thrift, careful saving